

PRAKTIKUM SISTEM OPERASI

MODUL 15

Keamanan



Oleh:

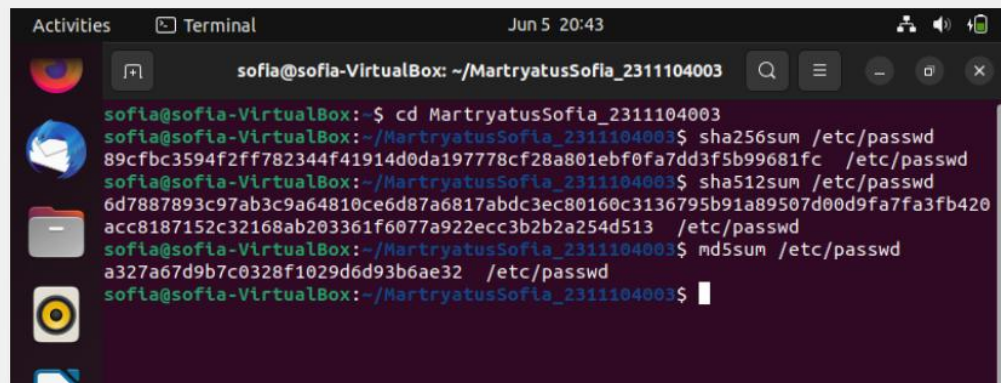
Martryatus Sofia

2311104003

JURNAL

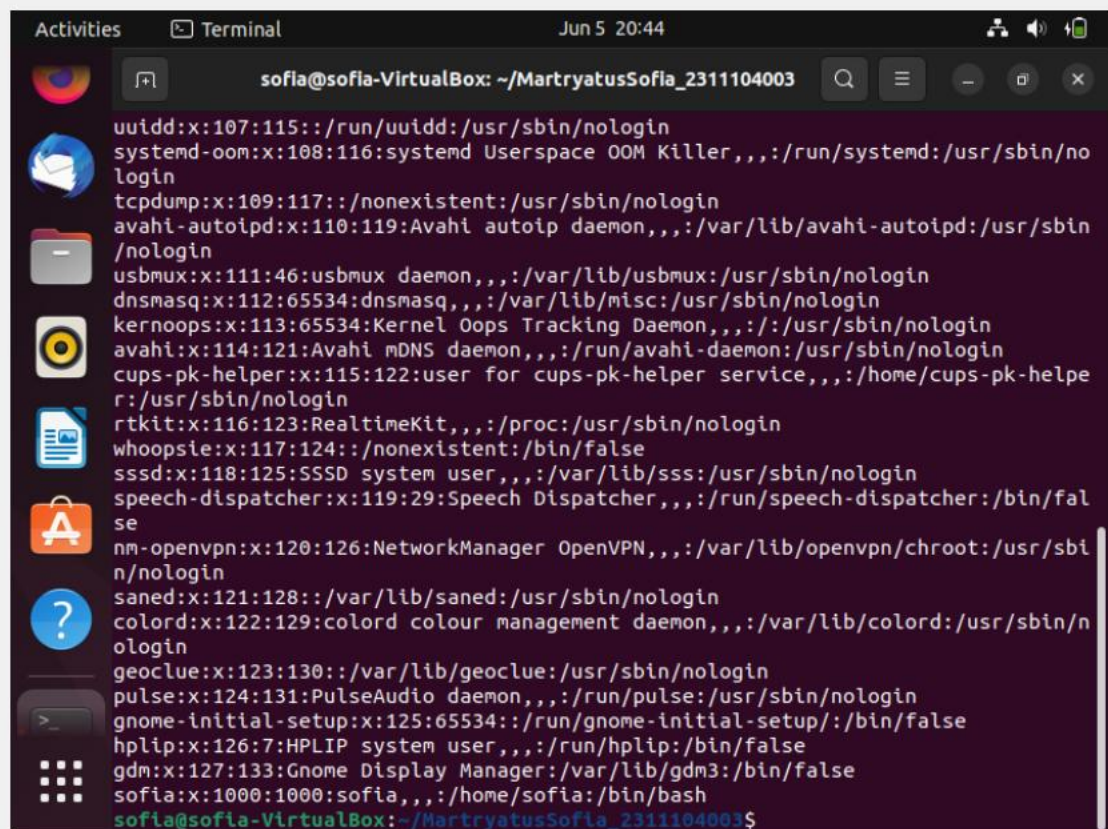
1. Integritas: dasar hashing [10 Point]

- a. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file /etc/passwd. Berapa nilai hash dari file /etc/passwd? Screenshot nilai hash dari file tersebut.



```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
sofia@sofia-VirtualBox:~$ cd MartryatusSofia_2311104003
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha256sum /etc/passwd
89cfbc3594f2ff782344f41914d0da197778cf28a801ebf0fa7dd3f5b99681fc /etc/passwd
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha512sum /etc/passwd
6d7887893c97ab3c9a64810ce6d87a6817abdc3ec80160c3136795b91a89507d00d9fa7fa3fb420
acc8187152c32168ab203361f6077a922ecc3b2b2a254d513 /etc/passwd
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ md5sum /etc/passwd
a327a67d9b7c0328f1029d6d93b6ae32 /etc/passwd
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

- b. [2 Point] Buatlah file bernama test_0.txt pada folder /home/praktikan. Isi file tersebut isi yang ada di file /etc/passwd (copy paste isi file /etc/passwd ke test_0.txt)



```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
uidd:x:107:115::/run/uidd:/usr/sbin/nologin
systemd-oom:x:108:116:systemd Userspace OOM Killer,,,:/run/systemd:/usr/sbin/nologin
tcpdump:x:109:117:./nonexistent:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
usbmux:x:111:46:usbmux daemon,,,:/var/lib/usbmux:/usr/sbin/nologin
dnsmasq:x:112:65534:dnsmasq,,,:/var/lib/misc:/usr/sbin/nologin
kernoops:x:113:65534:Kernel Oops Tracking Daemon,,,:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
rtkit:x:116:123:RealtimeKit,,,:/proc:/usr/sbin/nologin
whoopsie:x:117:124:./nonexistent:/bin/false
sssd:x:118:125:SSSD system user,,,:/var/lib/sss:/usr/sbin/nologin
speech-dispatcher:x:119:29:Speech Dispatcher,,,:/run/speech-dispatcher:/bin/false
nm-openvpn:x:120:126:NetworkManager OpenVPN,,,:/var/lib/openvpn/chroot:/usr/sbin/nologin
saned:x:121:128:./var/lib/saned:/usr/sbin/nologin
colord:x:122:129:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
geoclue:x:123:130:./var/lib/geoclue:/usr/sbin/nologin
pulse:x:124:131:PulseAudio daemon,,,:/run/pulse:/usr/sbin/nologin
gnome-initial-setup:x:125:65534:./run/gnome-initial-setup:/bin/false
hplip:x:126:7:HPLIP system user,,,:/run/hplip:/bin/false
gdm:x:127:133:Gnome Display Manager:/var/lib/gdm3:/bin/false
sofia:x:1000:1000:sofia,,,:/home/sofia:/bin/bash
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

- c. [2 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_0.txt. Berapa nilai hash dari file test_0.txt? Screenshot nilai hash dari file test_0.txt.

```
sofia:x:1000:1000:sofia,,,:/home/sofia:/bin/bash
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha256sum test_0.txt
89cfbc3594f2ff782344f41914d0da197778cf28a801ebf0fa7dd3f5b99681fc test_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha512sum test_0.txt
6d7887893c97ab3c9a64810ce6d87a6817abdc3ec80160c3136795b91a89507d00d9fa7fa3fb420
acc8187152c32168ab203361f6077a922ecc3b2b2a254d513 test_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ md5sum test_0.txt
a327a67d9b7c0328f1029d6d93b6ae32 test_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$
```

- d. [2 Point] Rename file test_0.txt menjadi file_0.txt. Lakukan hash SHA256, SHA512 dan MD5 untuk file_0.txt. Berapa nilai hash file_0.txt? Screenshot nilai hash dari file_0.txt.

```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ mv test_0.txt file_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha256sum file_0.txt
89cfbc3594f2ff782344f41914d0da197778cf28a801ebf0fa7dd3f5b99681fc file_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha512sum file_0.txt
6d7887893c97ab3c9a64810ce6d87a6817abdc3ec80160c3136795b91a89507d00d9fa7fa3fb420
acc8187152c32168ab203361f6077a922ecc3b2b2a254d513 file_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ md5sum file_0.txt
a327a67d9b7c0328f1029d6d93b6ae32 file_0.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$
```

- e. [2 Point] Apa hasil pengamatan Anda? File apa saja yang mempunyai hash yang sama? Jelaskan!

Jawab: File /etc/passwd, test_0.txt, dan file_0.txt memiliki nilai hash yang sama karena isi file tidak berubah. Perubahan nama file tidak memengaruhi nilai hash, karena hash dihitung berdasarkan isi file, bukan nama file.

2. Integritas: aavance [15 Point]

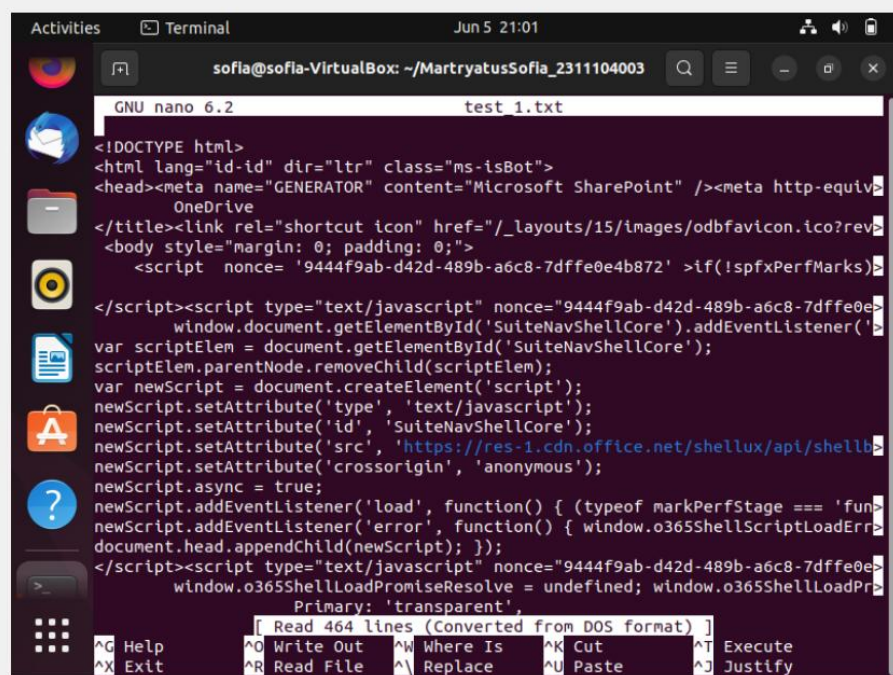
- a. [3 Point] Download file bernama test_1.txt di link ini tiny.cc/test1_txt

```
Activities Terminal Jun 5 20:49
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ wget http://tiny.cc/test1_
txt -O test_1.txt
--2026-06-05 20:49:36-- http://tiny.cc/test1_txt
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)[157.245.113.153]:80... connected.
HTTP request sent, awaiting response... 301 Moved Permanently
Location: https://tiny.cc/test1_txt [following]
--2026-06-05 20:49:37-- https://tiny.cc/test1_txt
Connecting to tiny.cc (tiny.cc)[157.245.113.153]:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:t:/g/personal/inf
ormaticslab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZlIGWnAcN25Y68gCPuWiGm_Rbm
coc [following]
--2026-06-05 20:49:38-- https://telkomuniversityofficial-my.sharepoint.com/:t:/
g/personal/infomaticslab_telkomuniversity_ac_id/IQDQY2UpngWaTb9EtnZlIGWnAcN25
Y68gCPuWiGm_Rbmccoc
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-
my.sharepoint.com)... 13.107.138.10, 13.107.136.10, 2620:1ec:8fa::10, ...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityoffic
ial-my.sharepoint.com)[13.107.138.10]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/infomat
icslab_telkomuniversity_ac_id/_layouts/15/onedrive.aspx?id=%2Fpersonal%2Finforma
ticslab%2Ftelkomuniversity%2FDocuments%2FIFLAB%20PERIOD%2025%2D26%2F
GANJIL%20FRAKTIKUM%2FIBANK%20SAL%2FISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUK%
2FIFILE%20DOWNLOA%2Ftest%2F1%2Etxt&parent=%2Fpersonal%2Finformatics%2Ftelkom
university%2FDocuments%2FIFLAB%20PERIOD%2025%2D26%2FGANJIL%2FRAKTIK
UM%2FIBANK%20SAL%2FISOP%2FMODUL%2014%20%2D%20KEAMANAN%20LINUK%2FIFILE%20DOWNLOA
D&ga=1 [following]
```

- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.txt. Berapa nilai hash test_1.txt? Screenshot nilai hash dari file test_1.txt.

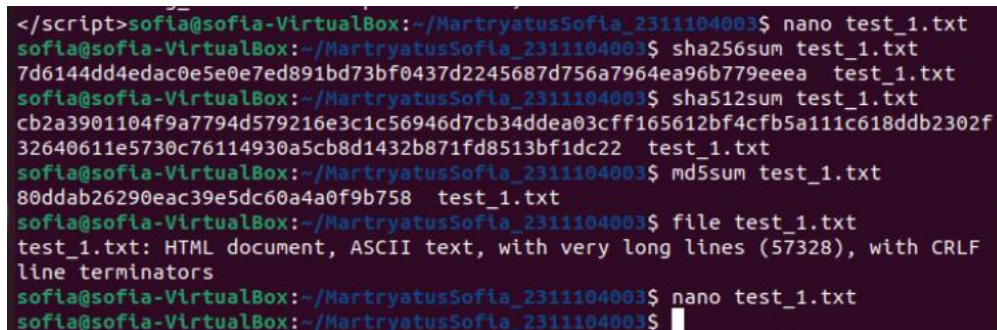
```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha256sum test_1.txt
dfc733950915042dd8c1442f357e27667abd69b73ff674a3974f9ce56b53c06f test_1.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ sha512sum test_1.txt
f4fadbd3e77b4d4577e624953833224751c791713169d413f94a7c2661554d0e80e0c6754979630
c1b072839b35cf00d5360d7dfe6e1a9d259b1da1370b402b4 test_1.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$ md5sum test_1.txt
5e3618e5f4d655498f167e4ca16690ec test_1.txt
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003$
```


- c. [3 Point] Hapuslah titik diakhir file test_1.txt tersebut, simpan file tersebut!



```
GNU nano 6.2 test_1.txt
<!DOCTYPE html>
<html lang="id-id" dir="ltr" class="ms-isBot">
<head><meta name="GENERATOR" content="Microsoft SharePoint" /><meta http-equiv=
OneDrive
</title><link rel="shortcut icon" href="/_layouts/15/images/odbfavicon.ico?rev=
<body style="margin: 0; padding: 0;">
<script nonce= '9444f9ab-d42d-489b-a6c8-7dffe0e4b872' >if(!spfXPerfMarks)>
</script><script type="text/javascript" nonce="9444f9ab-d42d-489b-a6c8-7dffe0e
window.document.getElementById('SuiteNavShellCore').addEventListener('
var scriptElem = document.getElementById('SuiteNavShellCore');
scriptElem.parentNode.removeChild(scriptElem);
var newScript = document.createElement('script');
newScript.setAttribute('type', 'text/javascript');
newScript.setAttribute('id', 'SuiteNavShellCore');
newScript.setAttribute('src', 'https://res-1.cdn.office.net/shellux/api/shellb
newScript.setAttribute('crossorigin', 'anonymous');
newScript.async = true;
newScript.addEventListener('load', function() { (typeof markPerfStage === 'fun
newScript.addEventListener('error', function() { window.o365ShellScriptLoadErr
document.head.appendChild(newScript); });
</script><script type="text/javascript" nonce="9444f9ab-d42d-489b-a6c8-7dffe0e
window.o365ShellLoadPromiseResolve = undefined; window.o365ShellLoadPr
Primary: 'transparent',
[ Read 464 lines (Converted from DOS format) ]
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^I Replace    ^U Paste      ^J Justify
```

- d. [3 Point] Lakukan hash dari SHA256, SHA512 dan MD5. Screenshot nilai hash dari file test_1.txt.



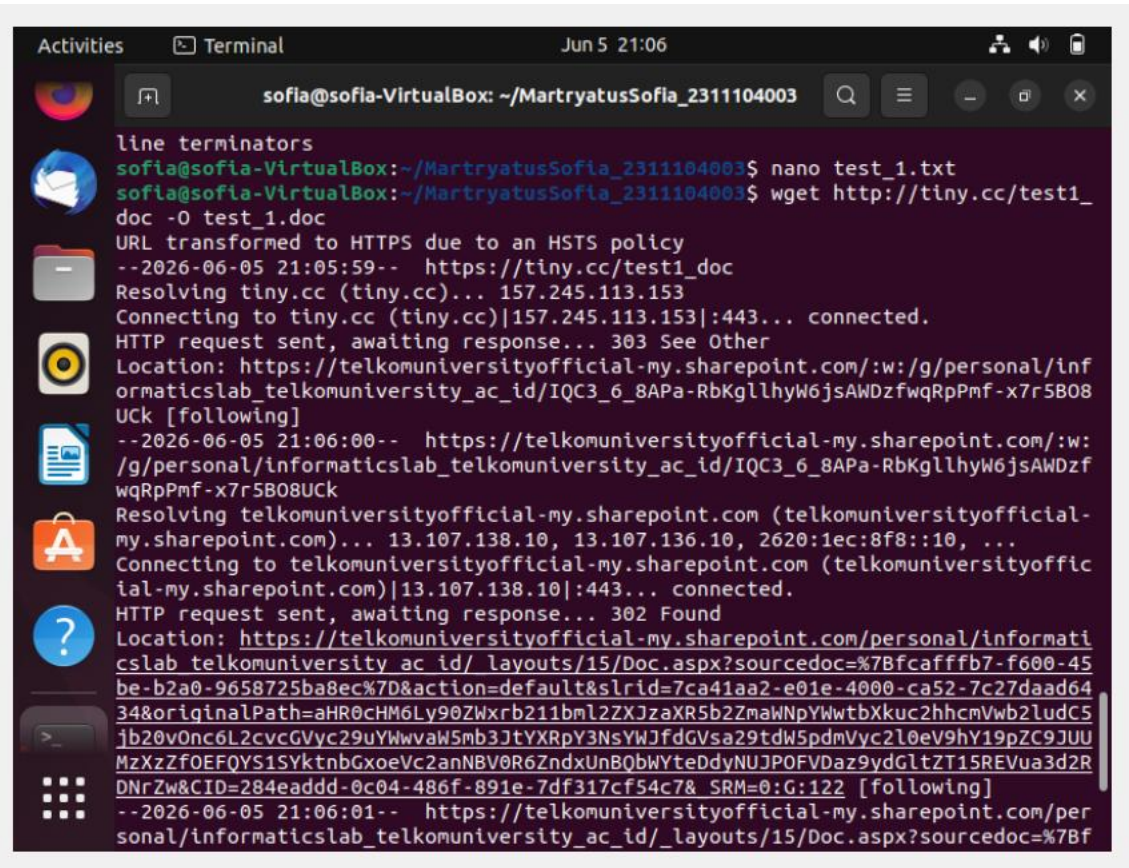
```
</script>sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ nano test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha256sum test_1.txt
7d6144dd4edac0e5e0e7ed891bd73bf0437d2245687d756a7964ea96b779eeaa test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha512sum test_1.txt
cb2a3901104f9a7794d579216e3c1c56946d7cb34ddea03cfff165612bf4cfb5a111c618ddb2302f
32640611e5730c76114930a5cb8d1432b871fd8513bf1dc22 test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ md5sum test_1.txt
80ddab26290eac39e5dc60a4a0f9b758 test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ file test_1.txt
test_1.txt: HTML document, ASCII text, with very long lines (57328), with CRLF
line terminators
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ nano test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

- e. [3 Point] Apa analisis (hasil pengamatan) Anda mengenai hal tersebut! Apakah nilai hash sama?

Jawab: Nilai hash berubah total meskipun hanya satu karakter yang dihapus. Hal ini menunjukkan sifat avalanche effect, yaitu perubahan kecil pada isi file akan menghasilkan nilai hash yang sangat berbeda.

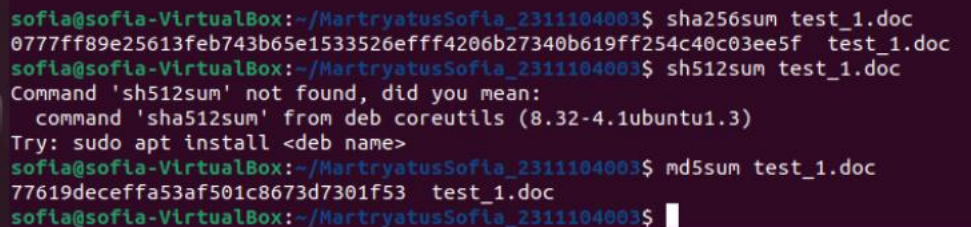
3. Integritas: metadata [18 Point]

- a. [3 Point] Download file bernama test_1.doc di link ini tiny.cc/test1_doc



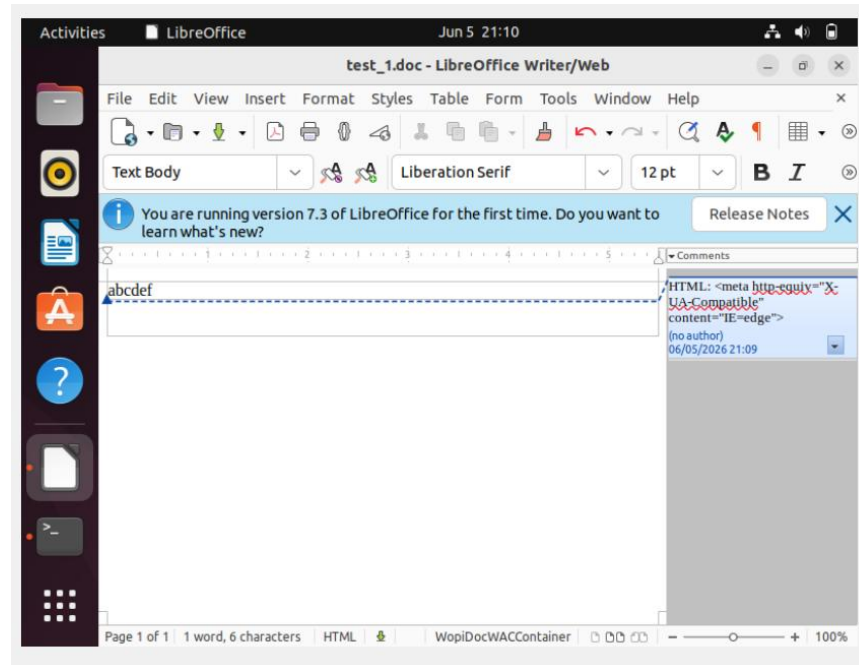
```
line terminators
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ nano test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ wget http://tiny.cc/test1_
doc -O test_1.doc
URL transformed to HTTPS due to an HSTS policy
--2026-06-05 21:05:59-- https://tiny.cc/test1_doc
Resolving tiny.cc (tiny.cc)... 157.245.113.153
Connecting to tiny.cc (tiny.cc)|157.245.113.153|:443... connected.
HTTP request sent, awaiting response... 303 See Other
Location: https://telkomuniversityofficial-my.sharepoint.com/:w:/g/personal/informatics_slab_telkomuniversity_ac_id/IQC3_6_8APa-RbKgllyW6jsAWDzfWqRpPmf-x7r5B08UCK [following]
--2026-06-05 21:06:00-- https://telkomuniversityofficial-my.sharepoint.com/:w:/g/personal/informatics_slab_telkomuniversity_ac_id/IQC3_6_8APa-RbKgllyW6jsAWDzfWqRpPmf-x7r5B08UCK
Resolving telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)... 13.107.138.10, 13.107.136.10, 2620:1ec:8f8::10, ...
Connecting to telkomuniversityofficial-my.sharepoint.com (telkomuniversityofficial-my.sharepoint.com)|13.107.138.10|:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://telkomuniversityofficial-my.sharepoint.com/personal/informatics_slab_telkomuniversity_ac_id/_layouts/15/Doc.aspx?sourcedoc=%7Bfcafff7-f600-45be-b2a0-9658725ba8ec%7D&action=default&slrid=7ca41aa2-e01e-4000-ca52-7c27daad6434&originalPath=aHR0cHM6Ly90ZWxrb211bml2ZXJzaXR5b2ZmaWNpYWwtbXkuc2hhcmVwb2ludC5jb20vOnc6L2cvGVyc29uYWwvaW5mb3JtYXRpY3NsYWJfdGVsa29tdW5pdmVyc2l0eV9hY19pZC9JUUMzXzZfOEFQYS1SYktnbGxoeVc2anNBV0R6ZndxUnB0bWYteDdyNUJPOFVDaz9ydGltZT1SREVua3d2RDNrZW&CID=284eadd-0c04-486f-891e-7df317cf54c7&SRM=0:G:122 [following]
--2026-06-05 21:06:01-- https://telkomuniversityofficial-my.sharepoint.com/personal/informatics_slab_telkomuniversity_ac_id/_layouts/15/Doc.aspx?sourcedoc=%7Bf
```

- b. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.

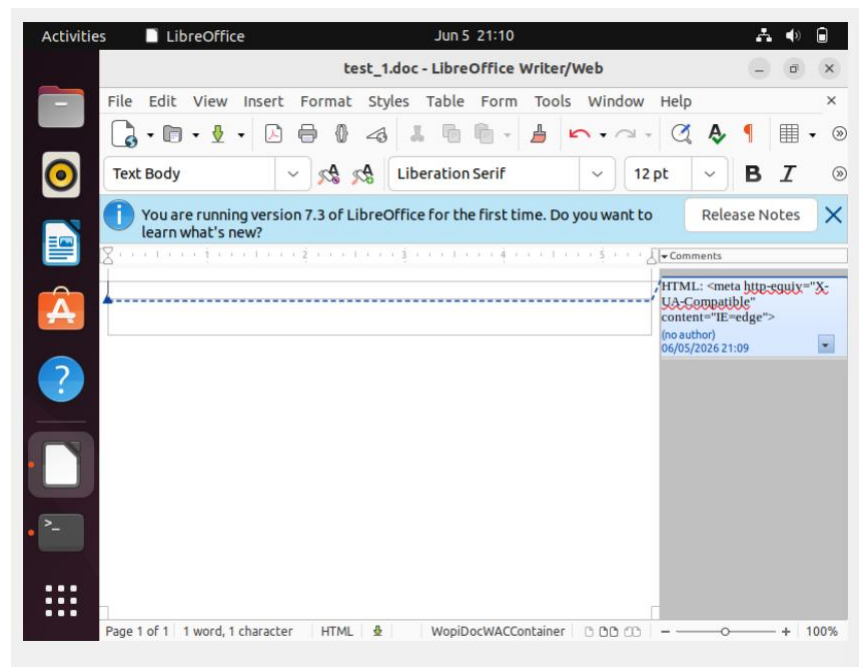


```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha256sum test_1.doc
0777ff89e25613feb743b65e1533526efff4206b27340b619ff254c40c03ee5f test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sh512sum test_1.doc
Command 'sh512sum' not found, did you mean:
  command 'sha512sum' from deb coreutils (8.32-4.1ubuntu1.3)
Try: sudo apt install <deb name>
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ md5sum test_1.doc
77619deceffa53af501c8673d7301f53 test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

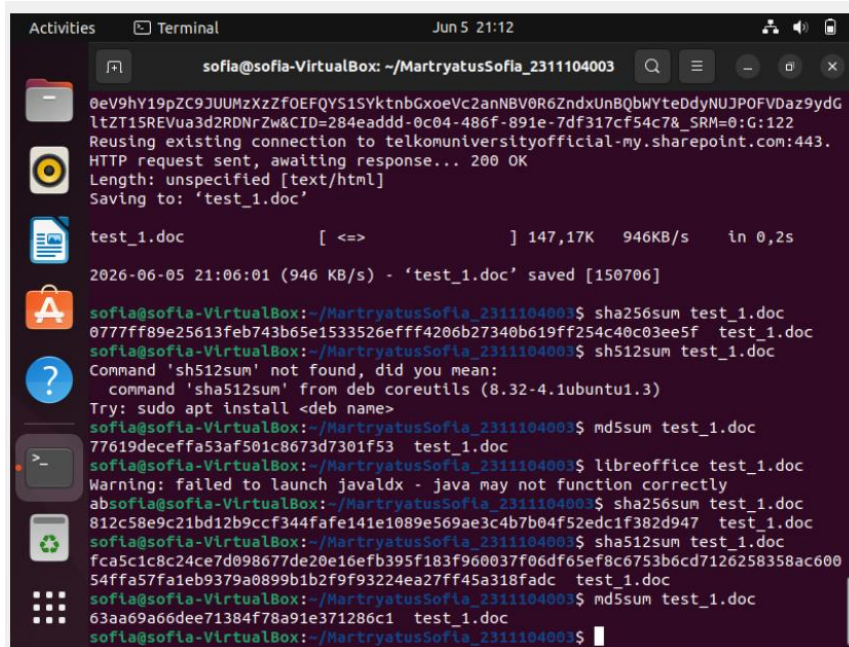
- c. Buka kembali file test_1.doc. Lakukan hal ini:
- [3 Point] Ketik abcdef. Save test_1.doc



ii. [3 Point] Hapus abcdef. Save test_1.doc



- d. [3 Point] Lakukan hash SHA256, SHA512 dan MD5 untuk file test_1.doc. Screenshot nilai hash dari file test_1.doc.



```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
0eV9hY19pZC9JUUMzXzZfOEFQYS1SYktnbGxoeVc2anNBV0R6ZndxUnBQbWYteDdyNUJPOFVDaz9ydG
ltZT15REvua3d2RDNrZW&CID=284eadd-0c04-486f-891e-7df317cf54c78_SRM=0:G:122
Reusing existing connection to telkomuniversityofficial-my.sharepoint.com:443.
HTTP request sent, awaiting response... 200 OK
Length: unspecified [text/html]
Saving to: 'test_1.doc'

test_1.doc      [ <=>      ] 147,17K  946KB/s   in 0,2s

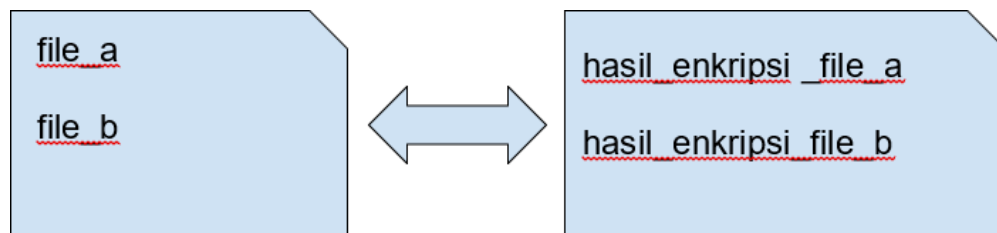
2026-06-05 21:06:01 (946 KB/s) - 'test_1.doc' saved [150706]

sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha256sum test_1.doc
0777ff89e25613feb743b65e1533526efff4206b27340b619ff254c40c03ee5f test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sh512sum test_1.doc
Command 'sh512sum' not found, did you mean:
  command 'sha512sum' from deb coreutils (8.32-4.1ubuntu1.3)
Try: sudo apt install <deb name>
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ md5sum test_1.doc
77619deceffa53af501c8673d7301f53 test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ libreoffice test_1.doc
Warning: failed to launch javaldx - java may not function correctly
absofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha256sum test_1.doc
812c58e9c21bd12b9ccf344fafe141e1089e569ae3c4b7b04f52edc1f382d947 test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ sha512sum test_1.doc
fca5c1c8c24ce7d098677de20e16efb395f183f960037f06df65ef8c6753b6cd7126258358ac600
54ffa57fa1eb9379a0899b1b2f9f9322ea27ff45a318fadc test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ md5sum test_1.doc
63aa69a66dee71384f78a91e371286c1 test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

- e. [3 Point] Hasil pengamatan apa yang diperoleh? Jelaskan alasannya!
Jawab: Nilai hash berubah walaupun isi dokumen dikembalikan seperti semula. Hal ini terjadi karena file dokumen menyimpan metadata, seperti waktu penyimpanan, riwayat perubahan, atau informasi aplikasi. Jadi, perubahan metadata juga memengaruhi nilai hash.

4. **Konfidensialitas: encfs [27 Point]**

Encfs adalah tool untuk melakukan mount dan membuat file system yang terenkripsi. Cara kerjanya adalah sebagai berikut. Ada 2 folder yang akan dibuat, dua folder tersebut saling terkait. Satu folder merupakan folder yang tidak terenkripsi, folder lainnya merupakan hasil enkripsi dari folder pertama.



WARNING: INSTALL ENCFs!

```
sudo apt-get update
sudo apt-get -y install encfs
```

Setelah terinstall silakan melanjutkan pengerjaan jurnal.

- a. [5 Point] Jalankan perintah berikut:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_normal
```

Pilih y (enter), pilih y (enter), enter, kemudian buatlah password. **Ingatlah password yang dibuat.** Setelah selesai, perhatikan bahwa telah muncul folder bernama folder_normal dan folder_terenkripsi.

```
Activities Terminal Jun 5 21:26
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003

The directory "/home/sofia/MartryatusSofia_2311104003/folder_normal/" does not
exist. Should it be created? (y,N) y
Creating new encrypted volume.
Please choose from one of the following options:
enter "x" for expert configuration mode,
enter "p" for pre-configured paranoia mode,
anything else, or an empty line will select standard mode.
?>

Standard configuration selected.

Configuration finished. The filesystem to be created has
the following properties:
Filesystem cipher: "ssl/aes", version 3:0:2
Filename encoding: "nameio/block", version 4:0:2
Key Size: 192 bits
Block Size: 1024 bytes
Each file contains 8 byte header with unique IV data.
Filenames encoded using IV chaining mode.
File holes passed through to ciphertext.

Now you will need to enter a password for your filesystem.
You will need to remember this password, as there is absolutely
no recovery mechanism. However, the password can be changed
later using encfstl.

New Encfs Password:
Verify Encfs Password:
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

- b. [5 Point] Copy dua atau tiga buah file (file apa saja) ke folder_normal. Amati dan tulis hasil observasi Anda pada folder_terenkripsi!

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ echo "file pertama" > fold
er_normal/file1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ echo "file kedua" > folder
_normal/file2.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ echo "file ketiga" > folde
r_normal/file3.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_normal
file1.txt file2.txt file3.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_terenkripsi
1Uj6irmXqgJ-jyhSSdJW147U Kuuy36eP-9ppfCm9bNgHB6fu RnPL1Ts-DNyGVuXz06gYWhHx
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

File yang dimasukkan ke folder_normal muncul di folder_terenkripsi dalam bentuk nama dan isi yang terenkripsi, sehingga tidak dapat dibaca secara langsung.

- c. [5 Point] Hapus salah satu file (bebas) pada folder_terenkripsi. Amati dan tulis hasil observasi Anda pada folder_normal!

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ rm folder_terenkripsi/MaC
ILE_TERENKRIPSI
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ rm folder_terenkripsi/RnPL
1Ts-DNyGVuXz06gYWhHx
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_normal
file1.txt file3.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

Ketika salah satu file terenkripsi dihapus, file aslinya pada folder_normal juga hilang karena kedua folder saling terhubung.

- d. [5 Point] Lakukan umount dengan perintah:

```
Fusermount -u ~/folder_anda/folder_normal
```

Amati dan tulis hasil observasi Anda pada folder_normal dan folder_terenkripsi


```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ fusermount -u ~/MartryatusSofia_2311104003/folder_normal
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_normal
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_terenkripsi
1Uj6irmXqgJ-jyhSSdJW147U Kuuy36eP-9ppfCm9bNgHB6fu
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

Setelah di-unmount, folder_normal menjadi kosong/tidak menampilkan file asli, sedangkan folder_terenkripsi tetap berisi file terenkripsi.

- e. [7 Point] Buatlah folder baru bernama folder_sembarang. Lakukan perintah berikut ini:

```
encfs ~/folder_anda/folder_terenkripsi ~/folder_anda/folder_sembarang
```

Amati dan tulis hasil observasi Anda!

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ encfs ~/MartryatusSofia_2311104003/folder_terenkripsi ~/MartryatusSofia_2311104003/folder_sembarang
EncFS Password:
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ encfs ~/MartryatusSofia_2311104003/folder_terenkripsi ~/MartryatusSofia_2311104003/folder_sembarang
EncFS Password:
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls
file_0.txt      folder_normal  folder_terenkripsi  test_1.txt
folder_laprak15 folder_sembarang test_1.doc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls folder_sembarang
file1.txt file3.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

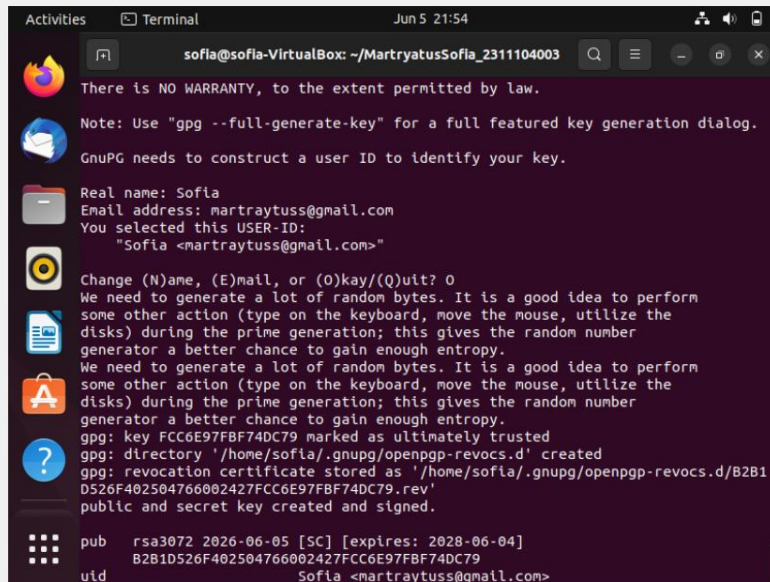
File asli dapat muncul kembali di folder_sembarang setelah folder terenkripsi di-mount menggunakan password yang benar.

5. Konfidensialitas: gpg [30 Point]

- a. [5 Point] Membuat kunci publik dan privat. Jalankan perintah ini:

```
gpg --gen-key
```

Baca dan ikuti perintah yang ada dari program gpg!



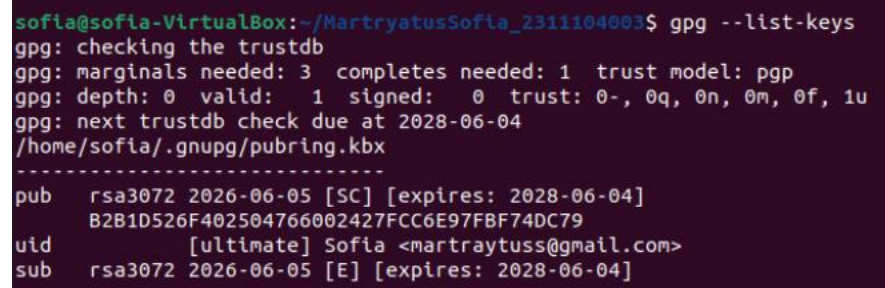
```
Activities Terminal Jun 5 21:54
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
There is NO WARRANTY, to the extent permitted by law.
Note: Use "gpg --full-generate-key" for a full featured key generation dialog.
GnuPG needs to construct a user ID to identify your key.
Real name: Sofia
Email address: martraytuss@gmail.com
You selected this USER-ID:
"Sofia <martraytuss@gmail.com>"
Change (N)ame, (E)mail, or (O)key/(Q)uit? 0
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
We need to generate a lot of random bytes. It is a good idea to perform
some other action (type on the keyboard, move the mouse, utilize the
disks) during the prime generation; this gives the random number
generator a better chance to gain enough entropy.
gpg: key FCC6E97FBF74DC79 marked as ultimately trusted
gpg: directory '/home/sofia/.gnupg/openpgp-revocs.d' created
gpg: revocation certificate stored as '/home/sofia/.gnupg/openpgp-revocs.d/B2B1
D526F402504766002427FCC6E97FBF74DC79.rev'
public and secret key created and signed.

pub  rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
     B2B1D526F402504766002427FCC6E97FBF74DC79
uid  Sofia <martraytuss@gmail.com>
```

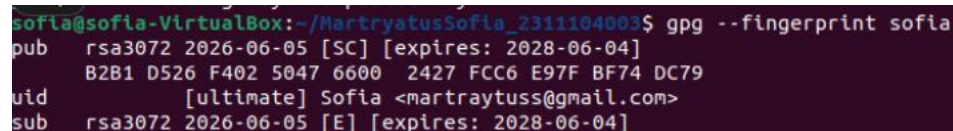
- b. [5 Point] Hasil publickey dan privatekey ada di folder ~/.gnupg. Privatekey tidak boleh keluar dari komputer ini dan hanya Anda saja yang dapat mengaksesnya. Kunci publik akan dibagikan kepada seluruh dunia atau untuk orang yang Anda inginkan saja. Kunci publik adalah pubring.gpg dan kunci private adalah secring.gpg. Lakukan perintah berikut ini untuk mengetahui daftar key yang Anda punya dan fingerprint kunci yang Anda punya!

```
gpg --list-keys
gpg --fingerprint nama@email.com
```

Screenshot hasil perintah di atas!



```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3 completes needed: 1 trust model: pgp
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u
gpg: next trustdb check due at 2028-06-04
/home/sofia/.gnupg/pubring.kbx
-----
pub  rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
     B2B1D526F402504766002427FCC6E97FBF74DC79
uid  [ultimate] Sofia <martraytuss@gmail.com>
sub  rsa3072 2026-06-05 [E] [expires: 2028-06-04]
```



```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --fingerprint sofia
pub  rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
     B2B1 D526 F402 5047 6600 2427 FCC6 E97F BF74 DC79
uid  [ultimate] Sofia <martraytuss@gmail.com>
sub  rsa3072 2026-06-05 [E] [expires: 2028-06-04]
```

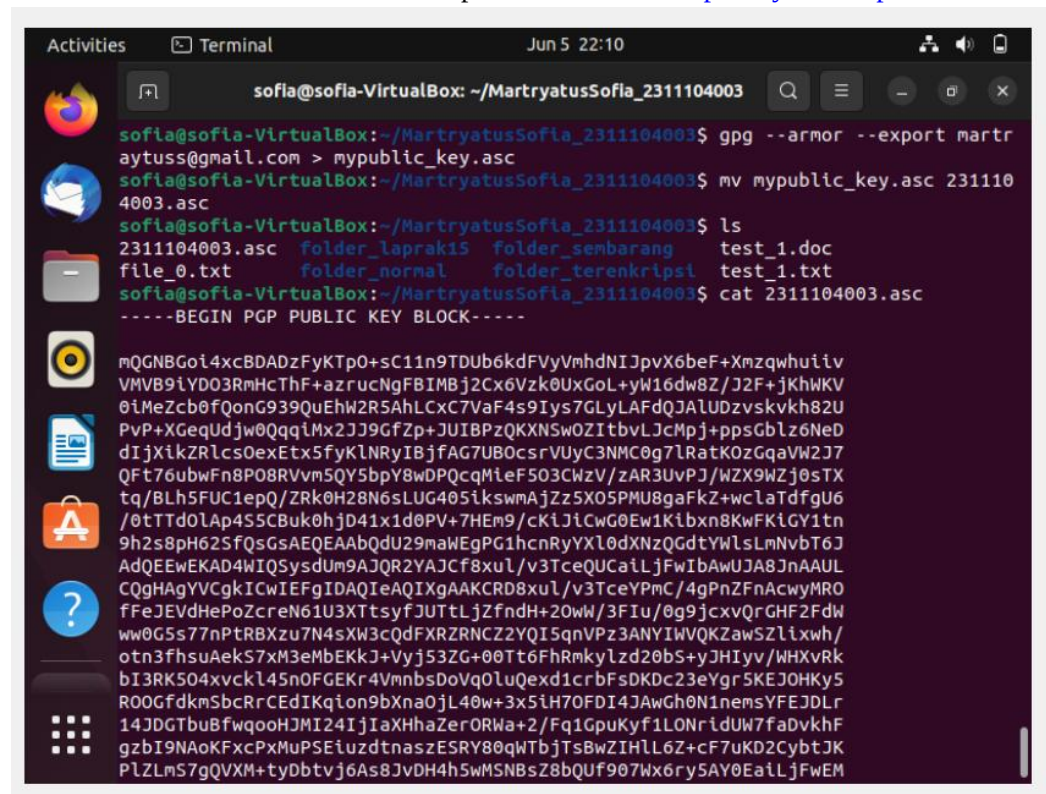
- c. [5 Point] Export kunci publik Anda. Jalankan perintah ini:

```
gpg --armor --export nama@email.anda > mypublic_key.asc
```

File **mypublic_key.asc** adalah file yang akan dibagikan kepada teman-teman Anda. Teman Anda akan menggunakan kunci publik Anda jika ingin mengirim pesan rahasia kepada Anda. Hanya Anda yang dapat membaca pesan tersebut karena hanya mempunyai kunci privat yang bersesuaian dengan kunci publik Anda.

- i. Rename mypublic_key.asc menjadi nim_anda.asc, Contoh: 130118xxxx.asc

- ii. Taruh file nim_anda.asc ke folder pada link berikut: <http://tiny.cc/SisopNomor5C>



```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --armor --export martr
aytuss@gmail.com > mypublic_key.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ mv mypublic_key.asc 231110
4003.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls
2311104003.asc  folder_laprak15  folder_sambarang  test_1.doc
file_0.txt       folder_normal    folder_terenkripsi test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ cat 2311104003.asc
-----BEGIN PGP PUBLIC KEY BLOCK-----
mQGNBGoI4xcBDADzFyKtP0+sC11n9TDUb6kdFVYVmhdlNIJpvX6beF+Xmqzwhuiiv
VMVB9iYD03RmHcThF+azrucNgFBIMBj2Cx6Vzk0UxGoL+yW16dw8Z/J2F+jKhWkv
0iMeZcb0fQonG939QuEhW2R5AhLCxC7VaF4s9Iys7GLyLAFdQJALUDzsvkvkh82U
PvP+XGeqUdjw0QqqIMx2JJ9GfZp+JUIBPzQKXNSWOZIbtvLJcMpj+ppsGblz6NeD
dIjXikZRLcsOexEtX5fyKlNRyIBjFAG7UB0csrVUyC3NMC0g7LRatK0zGqaVW2J7
QfT76ubwFn8P08RVm5QY5bpY8wDPQcmIEF503CwZV/zAR3UvPJ/WZX9WZj0sTX
tq/BLh5FUC1epQ/ZRk0H28N6sLUG405ikswwAjZz5X05PMU8gaFkZ+wclaTdfgU6
/0tTTd0LAp455CBuk0hjD41x1d0PV+7HEm9/cKlJlCwG0Ew1Kibxn8KwFKlGy1t
9h2s8pH625fQsGSAEQEAAbQdU29maHEGPG1hcnRyYXl0dXNzQgdTYWlsLmNvbT6J
AdQEwEKAD4WtQSYsdUm9AJQR2YAJCf8xul/v3TceQUcailjFwIbAwUJABJnAAUL
CQGHAgYVCgkICwIEFgIDAQIEAQIXgAAKCRD8xul/v3TceYpnc/4gPnZFnAcwyMRO
fFeJEVDHePoZcreN61U3XTtsyfJUTtLjZfndH+20wW/3FIu/0g9jcxvQrGHF2FdW
ww0G5s77nPtR8Xzu7N4sXW3cQdFXRZRNcZ2YQI5qnVPz3ANYIwVQKZawSZlixwh/
otn3fhsuAekS7xM3eMbEKkV+Vyj53ZG+00Tt6FhRmkyLzd20bS+yJHIyv/WHXvRk
bI3RK504xvckl45n0FGEKr4VmnbsDoVq0LuQexd1crbFSDKDC23eYgr5KEJ0HKy5
R00GfdkmsBcRrCEdIKqLon9bXna0jL40w+3x5iH70FDI4JAwGh0N1nemsYFEJDLr
14JDGTbuBfWqooHJMI24IjIaXHhaZerORWa+2/Fq1GpuKyf1LONridUW7faDvkHF
gzbI9NAoKfXcPxMuPSEiuzdtnaszESRY80qWtbjTSBwZIHLL6Z+cF7uKD2CytBJK
PLZLmS7gQVXM+tyDbtvj6As8JvDH4h5wMSNBsZ8BQUf907Wx6ry5AY0EaLjFwEM
```

- d. [5 Point] Mengimport kunci publik orang lain. Silakan download file nim_teman_sebelah_anda.asc dan jalankan perintah ini untuk mengimport (menambahkan kunci publik orang lain ke sistem Anda):

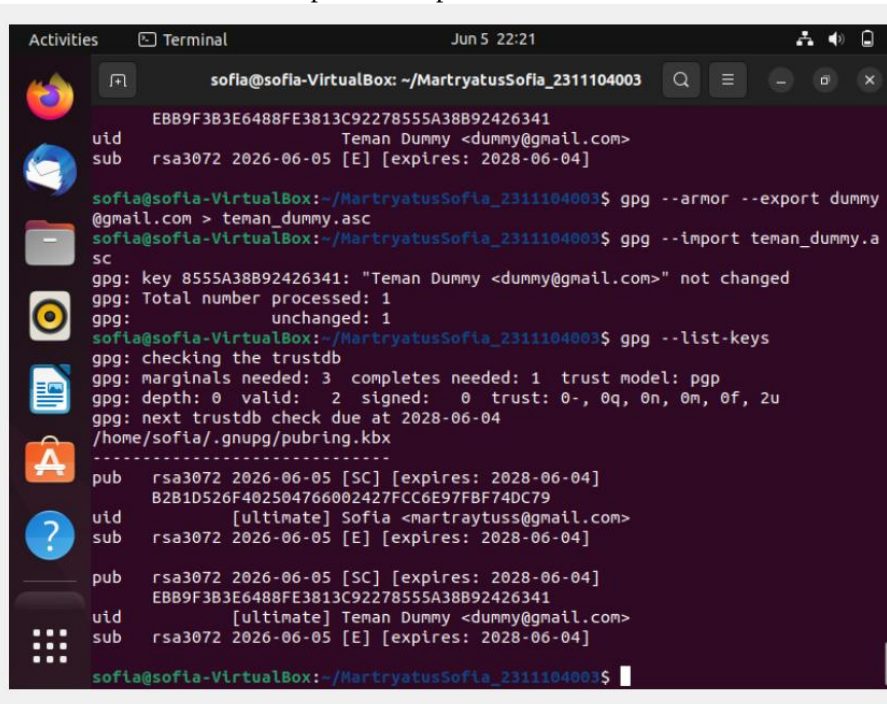
```
gpg --import nim_teman_sebelah_anda.asc
```

Contoh: “gpg --import 130118yyyy.asc” Setelah mengimport kunci publik teman Anda, Anda dapat mengirimkan pesan kepada teman Anda secara terenkripsi.

Untuk memastikan bahwa kunci publik telah diimport lakukan perintah

```
gpg --list-keys
```

dan nama teman Anda ada pada hasil perintah tersebut.



```
sofia@sofia-VirtualBox: ~/MartryatusSofia_2311104003
uid      EBB9F3B3E6488FE3813C9227855A38B92426341
sub      Teman Dummy <dummy@gmail.com>
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --armor --export dummy
@gmail.com > teman_dummy.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --import teman_dummy.a
sc
gpg: key 8555A38B92426341: "Teman Dummy <dummy@gmail.com>" not changed
gpg: Total number processed: 1
gpg:      unchanged: 1
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --list-keys
gpg: checking the trustdb
gpg: marginals needed: 3  completes needed: 1  trust model: pgp
gpg: depth: 0  valid: 2  signed: 0  trust: 0-, 0q, 0n, 0m, 0f, 2u
gpg: next trustdb check due at 2028-06-04
/home/sofia/.gnupg/pubring.kbx
-----
pub      rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
          B2B1D526F402504766002427FCC6E97FBF74DC79
          [ultimate] Sofia <martraytuss@gmail.com>
uid      rsa3072 2026-06-05 [E] [expires: 2028-06-04]
sub      EBB9F3B3E6488FE3813C9227855A38B92426341
          [ultimate] Teman Dummy <dummy@gmail.com>
pub      rsa3072 2026-06-05 [SC] [expires: 2028-06-04]
uid      EBB9F3B3E6488FE3813C9227855A38B92426341
sub      [ultimate] Teman Dummy <dummy@gmail.com>
uid      rsa3072 2026-06-05 [E] [expires: 2028-06-04]
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```


- e. **[5 Point] Enkripsi pesan.** Buatlah file bernama file_rahasia.txt. Isi file tersebut dengan pesan rahasia Anda. Pesan inilah yang akan Anda kirim ke teman Anda. Jalankan perintah ini untuk melakukan enkripsi:

```
//ditulis dalam satu baris
gpg --encrypt --armor -r
alamat_email_teman_anda_yang_baru_diimport@xxx.com file_rahasia.txt
```

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ nano file_rahasia.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg --encrypt --armor -r d
ummy@gmail.com file_rahasia.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls
2311104003.asc  file_rahasia.txt.asc  folder_senbarang  test_1.doc
file_0.txt      folder_laprak15      folder_terenkripsi  test_1.txt
file_rahasia.txt  folder_normal      teman_dummy.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ mv file_rahasia.txt.asc fi
le_rahasia_untuk_dummy.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls *.asc
2311104003.asc  file_rahasia_untuk_dummy.asc  teman_dummy.asc
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```

```
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ ls -la
total 396
drwxrwxr-x  6 sofia sofia  4096 Jun  5 22:30 .
drwxr-x--- 19 sofia sofia  4096 Jun  5 20:49 ..
-rw-rw-r--  1 sofia sofia  2448 Jun  5 22:08 2311104003.asc
-rw-r--r--  1 sofia sofia  2810 Jun  5 20:44 file_0.txt
-rw-rw-r--  1 sofia sofia    25 Jun  5 22:24 file_rahasia.txt
-rw-rw-r--  1 sofia sofia   736 Jun  5 22:25 file_rahasia_untuk_dummy.asc
drwxrwxr-x  2 sofia sofia  4096 Jun  5 21:41 folder_laprak15
drwx----- 2 sofia sofia  4096 Jun  5 21:25 folder_normal
drwx----- 2 sofia sofia  4096 Jun  5 21:33 folder_senbarang
drwx----- 2 sofia sofia  4096 Jun  5 21:33 folder_terenkripsi
-rw-rw-r--  1 sofia sofia  2448 Jun  5 22:19 teman_dummy.asc
-rw-rw-r--  1 sofia sofia   779 Jun  5 21:10 test_1.doc
-rw-rw-r--  1 sofia sofia 353863 Jun  5 20:58 test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg -d file_rahasia_untuk_
```

- f. **[5 Point] Hanya teman anda yang dapat membuka file tersebut.** Hasil dari proses tersebut adalah file_rahasia_untuk_teman_anda.asc (contoh: file_rahasia_untuk_130118yyyyy.asc) Taruh file_rahasia_untuk_teman_anda.asc ke folder pada link:

<http://tiny.cc/SisopNomor5F>

Download dan buka file yang diperuntukkan bagi Anda dan lakukan dekripsi untuk melihat isi pesan dengan perintah:

```
gpg file_rahasia_nim_anda.asc
```

```
-rw-rw-r--  1 sofia sofia 353863 Jun  5 20:58 test_1.txt
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$ gpg -d file_rahasia_untuk_
dummy.asc
gpg: encrypted with 3072-bit RSA key, ID 238EA959B0BFC89E, created 2026-06-05
      "Teman Dummy <dummy@gmail.com>"
Hallloooooo temannn kecil
sofia@sofia-VirtualBox:~/MartryatusSofia_2311104003$
```